



TECHNISCHE UNIVERSITÄT
BERGAKADEMIE FREIBERG

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A report on

Image convolution with MPI

Programming Project | Winter Semester 2020-2021

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March 5, 2021

Eidesstattliche Erklärung

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5. März 2021

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Declaration

I hereby declare that I completed this work without any improper help from a third party and without using any aids other than those cited. All ideas derived directly or indirectly from other sources are identified as such. This declaration also refers to the representation of figures and visual material.

March 5, 2021

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Contents

1	Introduction	5
2	Convolution kernels	5
2.1	Gaussian Blur	5
2.2	Sharpen	6
2.3	Edge Detection	6
3	MPI Parallelization and Communication	7
4	Results	7
4.1	8162 pixel image	7
4.2	512 pixel image	8

List of Figures

1	Distribution of data between processes	7
2	512 x 512 grayscale image	8

List of Tables

1	Table of times computed on different processes.	7
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1 Introduction

This programming project focuses on convolution of a grayscale image with the following three convolution kernels:

- Blurring
- Sharpening
- Edge Detection

The program thus developed has been implemented in parallel with Message Passing Interface (MPI) using C++ programming language in the high performance computing cluster of TU Bergakademie Freiberg. The program has been implemented on the following two images:

- 512 x 512 pixel grayscale image.
- 8162 x 8162 pixel grayscale image.

No suited external libraries have been used or implemented in the code belonging to this project.

Format of the grayscale image

The images provided are in Netpbm grayscale image format. Quoted as "the lowest common denominator grayscale file format," this format had been chosen due to its ease in coding. Unlike the color images consisting of three color channels, with every pixel displaying the combined information of Red, Green and Blue channels, this format consists of one channel. The images considered in this project are said to be of 8-bit depth, i.e. the values of the pixels lie within the interval $[0, 255]$. The value 0 describes black, whereas 255 describes white. The images provided within this project are of uncompressed PGM data format. Therefore, initially, these values are read into a one-dimensional array, which are worked upon. Upon completion, the resultant array are written to file as per the specifications, which can then be converted to different formats using ImageMagick application.

2 Convolution kernels

Within the context of image processing, a convolution kernel, or a kernel, refers to a matrix that allows manipulation of values of pixel as per requirements. The procedure, known as convolution, refers to the process of computing a new pixel value from the given pixel value along with its surrounding pixel values. The kernels are specified below.

2.1 Gaussian Blur

The aim of this kernel is to smoothen an image, or reduce the noise or detail within an image. The 3 x 3 matrix for this operation is specified below.

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

2.2 Sharpen

The aim of this kernel is to enhance the sharpness of the image, can be understood to be the opposite of blur. The 3 x 3 matrix for this operation is specified below.

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

2.3 Edge Detection

The aim of this kernel is to identify the edges present in the image. In the context of image processing, an edge can be understood as the collection of points where the brightness suddenly changes, or has discontinuities. The 3 x 3 matrix used for this operation is shown below.

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Then, the respective convolution kernel is then applied to a pixel value $I(x,y)$ to obtain a new pixel value $I^*(x,y)$ where (x,y) denotes the position in the image. Convolution procedure can be shown in a mathematical format as follows.

$$I^*(x,y) = \sum_{i=1}^n \sum_{j=1}^n I(x-i+2, y-j+2)k(i,j)$$

where k represents the chosen 3 x 3 convolution kernel. An important aspect to note is the treatment of pixel values at the boundaries of the image. In this implementation, these are treated periodically, i.e. the pixel value at position $(1,0)$ receives information from $(1,511)$, the opposite border, in order to finish the computation of new pixel value. This is achieved by using modulo operator within the loops, as opposed to another popular method known as padding.

3 MPI Parallelization and Communication

For the parallel implementation of this code, MPI library is applied. The pixel values are thus distributed in a way such that each process has its own chunk of rows of pixel values, including the information belonging to the halo. Here, a halo refers to the rows of pixels indicated by dashed lines in the figure. The figure representing this division can be seen here. Subsequent execution of this program is achieved using university cluster's PBS queue system, using a script file. The wall times are then tabulated.

4 Results

4.1 8162 pixel image

The image of 8162 x 8162 dimensions is parallelized using Message Passing Interface (MPI) and executed on the HPC computing clusters of TU Bergakademie Freiberg. The executions are timed and coordinated using PBS Queuing system as designated by the university. Wall time calculated using the function are tabulated below:

`MPI_Wtime()`

#P	Time
1	110.34
2	65.75
4	44.18
6	38.5
8	35.18
12	32.76
24	53.12
36	131

Tab. 1: Table of times computed on different processes.

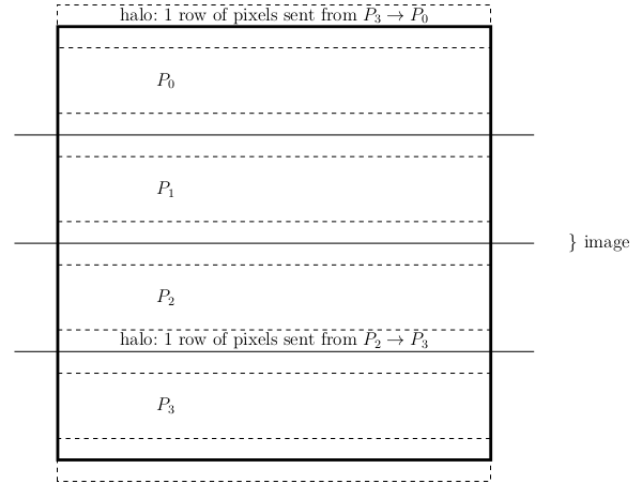


Fig. 1: Distribution of data between processes

4.2 512 pixel image

At the beginning, the results belonging to the 512 x 512 image are presented.

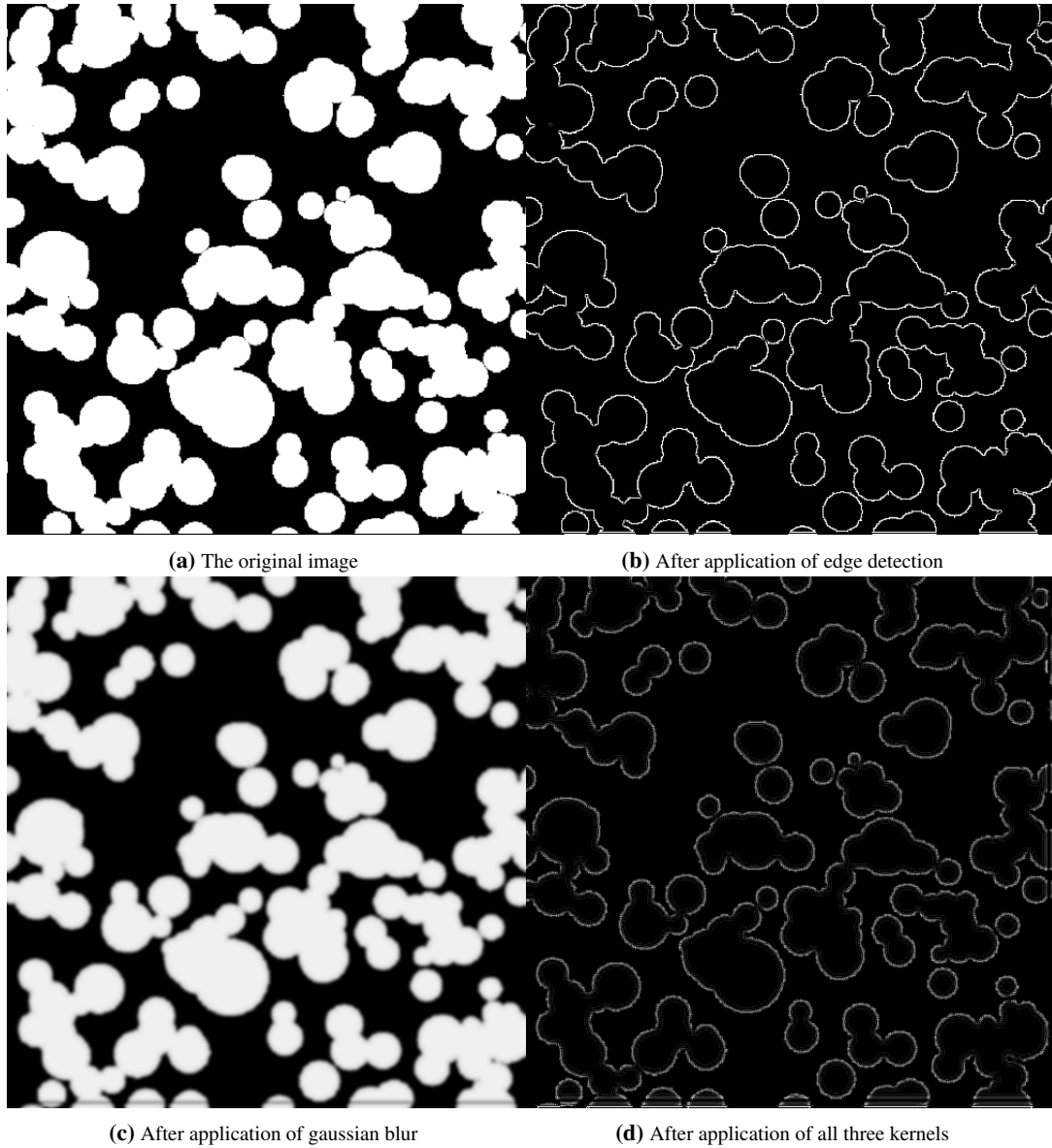


Fig. 2: 512 x 512 grayscale image

Appendix

PBS Script

The PBS script used to execute the program on the High performance computing cluster of the university is shown below.

```
#!/bin/bash
#PBS -N MPI
#PBS -q teaching
#PBS -l select=1:ncpus=1:mem=2gb:mpiprocs=1
#PBS -l walltime=00:03:00
#PBS -o pbs_out.txt
#PBS -e pbs_err.txt
module load gnu/gcc/8.1.0
module load openmpi/gcc/7.1.0/2.1.2
PBS_O_WORKDIR=$HOME/project/
cd $PBS_O_WORKDIR
cat $PBS_NODEFILE > pbs_info.txt
echo $OMP_NUM_THREADS >> pbs_info.txt
mpic++ -o main.out main.cpp
mpirun ./main.out
```